D2.1 Cell and nuclear division

Notes written for this site. Slides, videos, worksheets and self-check questions: whatthecell.org/subtopics/sl-d2-1.html

Written for this site from the course textbook and class slides, with page references for each statement. The chapter is on pages 648–661 of the course textbook. Image credits are on the last slide of the slides for this subtopic.

Why cells divide

D2.1.1 Making new cells

p. 648

A parent cell, often called a mother cell, divides to form two daughter cells. New cells arise only by division of existing cells. Division provides for growth, repair, maintenance and reproduction.

D2.1.4 What mitosis and meiosis each do

pp. 650–651

Mitosis maintains the chromosome number and the genome. **Meiosis** halves the chromosome number, and generates genetic diversity. The nucleus divides before the cell, so that no anucleate cells are produced.

D2.1.5 Why DNA is replicated first

p. 651

Replication in interphase gives each chromosome two DNA molecules, called **chromatids**, held together at a **centromere**. Each new nucleus can then receive a full set of genes.

Watch out: a replicated chromosome still counts as one chromosome. Once chromatids separate, each is counted as a chromosome.

D2.1.6 Condensation and movement of chromosomes

pp. 652–653

DNA wraps around proteins called **histones**, and further coiling, known as **supercoiling**, gives short, compact chromosomes. **Microtubules** form a spindle, and microtubule motors use ATP to move the chromosomes. Both mitosis and meiosis share these features.

Watch out: centromere, centrosome and centriole are three different structures.

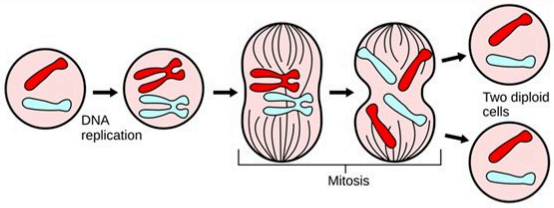
Mitosis and cytokinesis

D2.1.7 The phases of mitosis

pp. 653–654

The phases of mitosis

D2.1.7



Major events in mitosis. Mysid, public domain.

Phase	What happens
Prophase	Chromosomes condense; nuclear membrane breaks down; spindle forms
Metaphase	Chromosomes line up at the equator; spindle attached to each centromere
Anaphase	Sister chromatids separate and move to opposite poles
Telophase	Nuclear membranes reform; chromosomes uncoil

D2.1 Cell and nuclear division – Textbook pp. 653–654 12

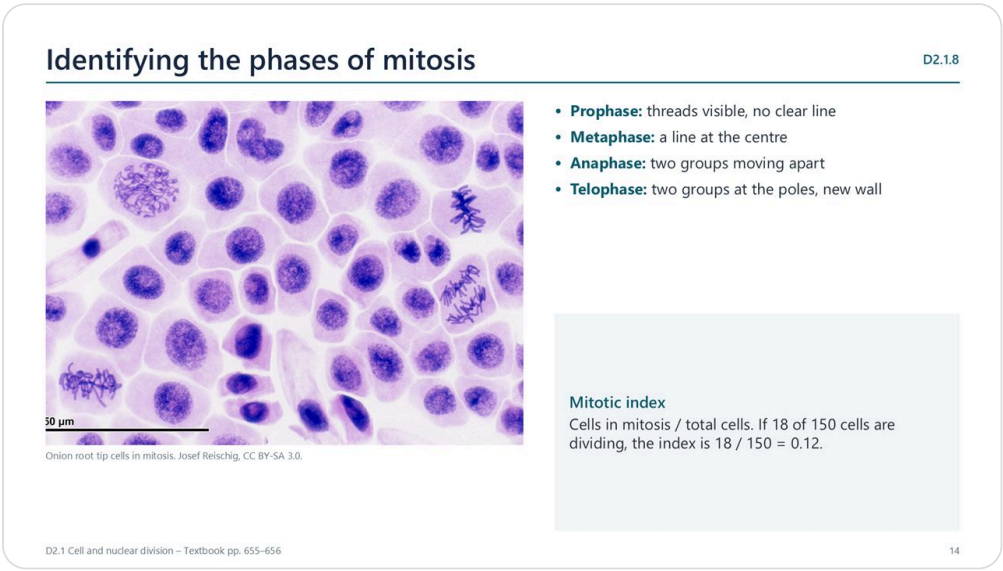
Mitosis gives two identical nuclei. DNA is copied exactly, and then one chromatid of each chromosome moves to each pole.

D2.1.8 Identifying the phases

pp. 655–656

- **Prophase:** threads visible, no clear line
- **Metaphase:** a line at the centre
- **Anaphase:** two groups moving apart, in V shapes
- **Telophase:** two groups at the poles

The **mitotic index** is the number of cells in mitosis divided by the total number of cells.



D2.1.2 Cytokinesis

p. 649

In animal cells, a ring of **actin and myosin** contracts and pinches the membrane together. In plant cells, **vesicles** gather at the equator and assemble sections of membrane and cell wall.

D2.1.3 Equal and unequal cytokinesis

pp. 649–650

Cytoplasm is usually shared equally. In **oogenesis** in humans, the egg keeps most of it, and in **yeast budding** a small bud forms. Each daughter cell needs at least one mitochondrion, and any other organelle made only by dividing an existing one.

Meiosis

D2.1.9 Meiosis halves the chromosome number

pp. 656–658

Diploid nuclei have two sets of chromosomes, in homologous pairs, and **haploid** nuclei have one set. One diploid nucleus gives four haploid nuclei. Sexual life cycles need this, since fertilization doubles the number.

	Meiosis I	Meiosis II
Separates	Homologous chromosomes	Sister chromatids
Result	Two haploid nuclei	Four haploid nuclei

D2.1.10 Non-disjunction and Down syndrome

pp. 658–659

In **non-disjunction**, chromosomes fail to separate in meiosis. If a gamete with two copies of chromosome 21 is fertilized, the zygote has three copies, and 47 chromosomes in total. This is trisomy 21, or **Down syndrome**. The risk rises with the age of the mother.

D2.1.11 How meiosis generates variation

pp. 660–661

Random orientation: each bivalent lines up independently in metaphase I. With a haploid number of n, there are 2 to the power n combinations, which is over 8 million in humans.

Crossing over: in prophase I, non-sister chromatids swap sections, giving new combinations of alleles on a chromosome.

Watch out: neither process creates new alleles. Only mutation does that.

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